```
#data
import pandas as pd
import matplotlib.pyplot as plt
data={
    "Sprint" :["S1", "S2", "S3"],
    "Task_completed":[25,45,78],
    "Bugs_reported":[1,2,6],
    "Hours_worked":[230,122,320],
    "Developers":["Team A","Team B","Team c"],
    "Code_coverage":[40,70,90]
}

df=pd.DataFrame(data)
print(df)
```

```
  Sprint  Task_completed  Bugs_reported  Hours_worked Developers  \
0     S1              25              1           230     Team A
1     S2              45              2           122     Team B
2     S3              78              6           320     Team c

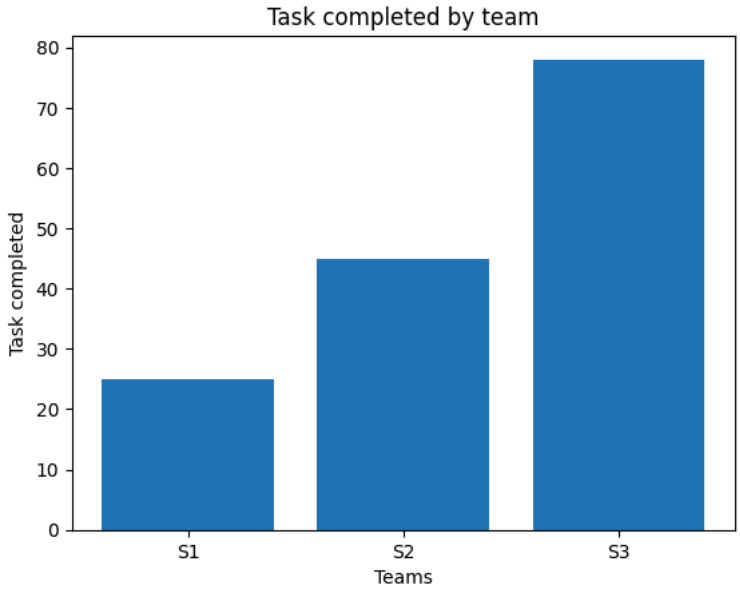
   Code_coverage
0             40
1             70
2             90
```

```
df.shape
```

```
(3, 6)
```

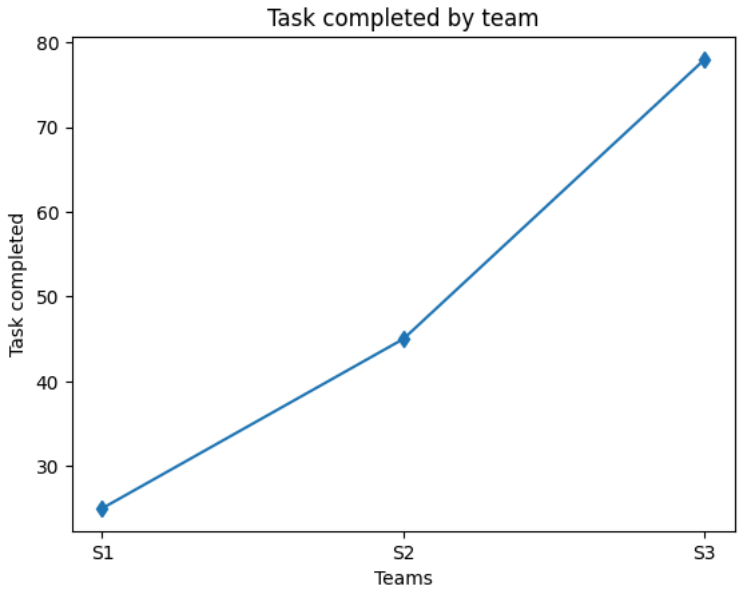
```
#Bar-graph

plt.bar(df["Sprint"], df["Task_completed"])
plt.title("Task completed by team")
plt.xlabel("Teams")
plt.ylabel("Task completed")
plt.show()
```



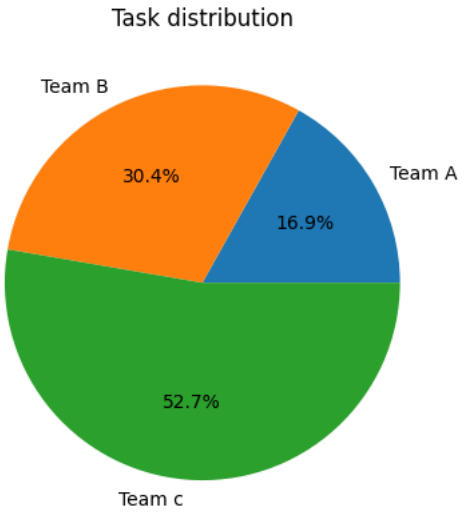

```
#Plot_graph

plt.plot(df["Sprint"], df["Task_completed"], marker="d")
plt.title("Task completed by team")
plt.xlabel("Teams")
plt.ylabel("Task completed")
plt.show()
```



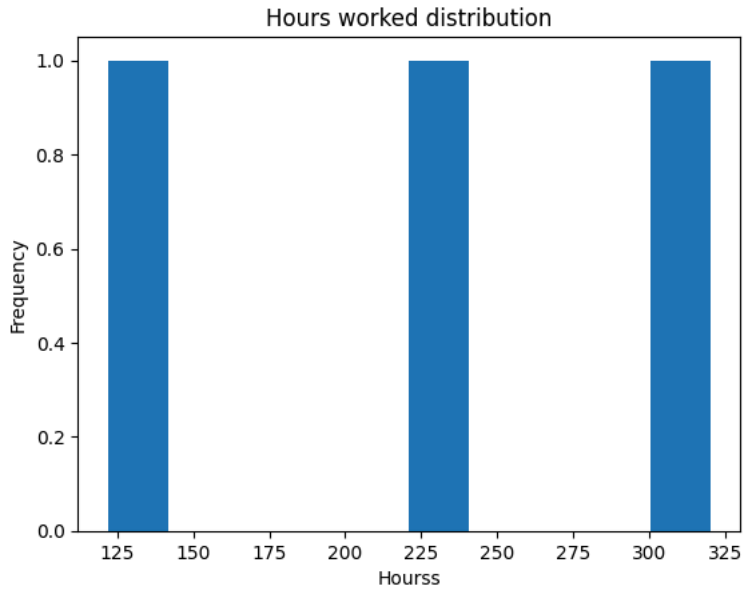

```
#Pie_chart

plt.pie(df["Task_completed"], labels=df["Developers"],autopct="%1.1f%%")
plt.title("Task distribution")
plt.show()
```



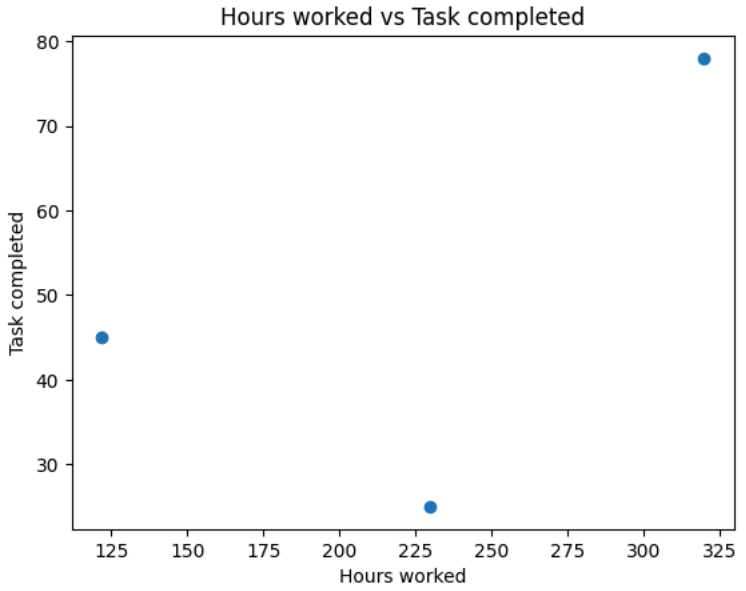

```
#histogram


plt.hist(df["Hours_worked"], bins=10)
plt.title("Hours worked distribution")
plt.xlabel("Hourss")
plt.ylabel("Frequency")
plt.show()
```



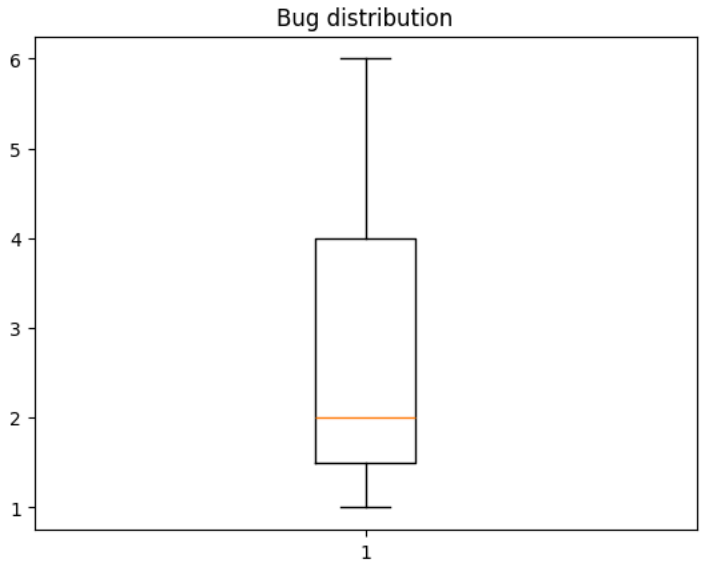

```
#scatter


plt.scatter(df["Hours_worked"], df["Task_completed"])
plt.title("Hours worked vs Task completed")
plt.xlabel("Hours worked")
plt.ylabel("Task completed")
plt.show()
```



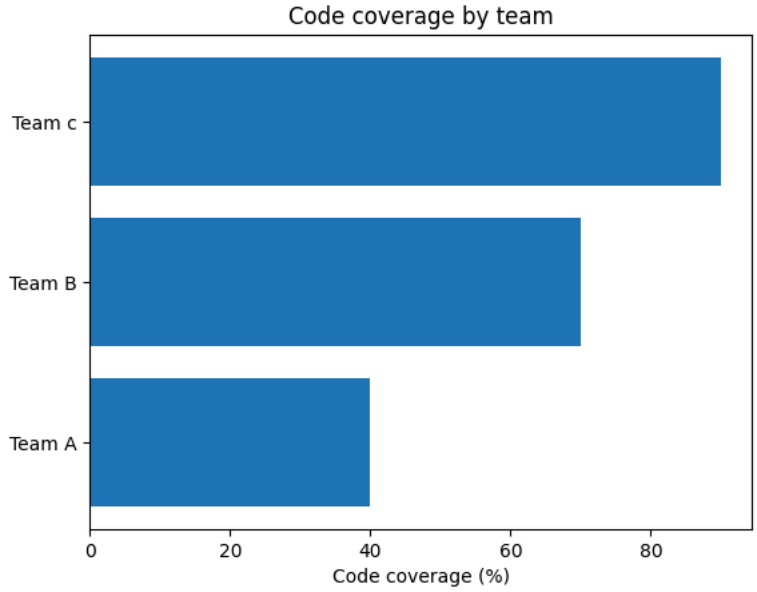

```
#box


plt.boxplot(df["Bugs_reported"])
plt.title("Bug distribution")
plt.show()
```



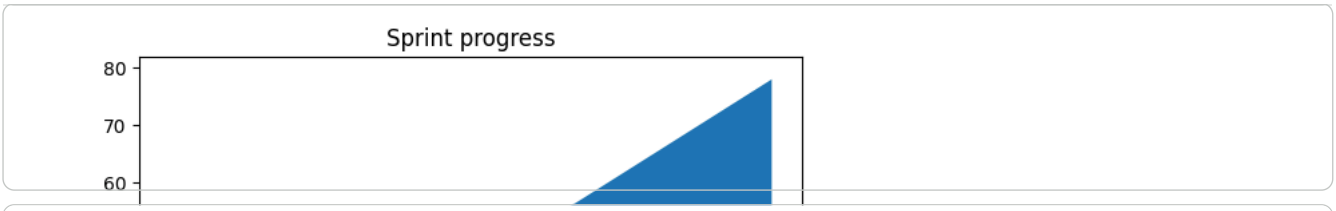

```
#barh_graph

plt.barh(df["Developers"], df["Code_coverage"])
plt.title("Code coverage by team")
plt.xlabel("Code coverage (%)")
plt.show()
```




```
#fill_between

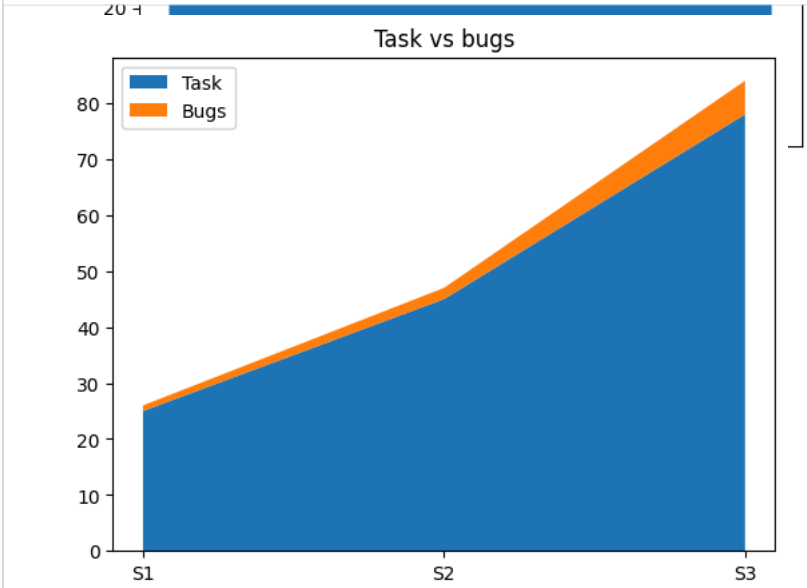
plt.fill_between(df["Sprint"],df["Task_completed"])
plt.title("Sprint progress")
plt.xlabel("Sprint")
plt.ylabel("Task commpleted")
plt.show()
```




```
#Stackplot

plt.stackplot(df["Sprint"],df["Task_completed"],df["Bugs_reported"],labels=["Task","Bugs"])

plt.legend()
plt.title("Task vs bugs")
plt.show()
```




```
#Project Metrics Heatmap

data = [df["Task_completed"],df["Bugs_reported"],df["Hours_worked"],df["Code_coverage"] ]

plt.imshow(data)
plt.xticks(range(3), df["Sprint"])
plt.yticks(range(4), ["Tasks", "Bugs", "Hours", "Coverage"])
plt.colorbar()
plt.title("Project Metrics Heatmap")
plt.show()
```